

Joshua Sakthivel Raju

AI/ML Research Engineer

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SUMMARY

Passionate AI/ML engineer with a strong background in deep learning, computer vision, and natural language processing. Skilled in Python, TensorFlow, PyTorch, and various ML libraries. Demonstrates excellent problem-solving, research, collaboration, and leadership abilities with unwavering work ethics and commitment to excellence. Proven track record of leading teams and projects to success. Seeking a challenging role to develop cutting-edge AI solutions.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, SQL, React.JS, HTML, SCSS

Technologies: Deep Learning, Computer Vision, Machine Learning, Natural Language Processing, 3D Modeling & Graphics, Web Development, Game Development

PROJECTS

Surgical Tool Segmentation

Python, TensorFlow, OpenCV

Deep Learning, Computer Vision

- Engineered advanced U-Net models that achieved a remarkable Intersection over Union (IoU) score of 94.8%, leading to substantial advancements in both surgical precision and team workflow effectiveness.

InTellik

React, JavaScript, Python

Natural Language Processing, Web Application

- Winner of Dataset 2024 National-Level Hackathon with more than 300+ participants.
- Fine-tuned Llama 3.1 (8B parameters) and implemented Retrieval-Augmented Generation (RAG) to leverage customer interaction histories for personalized service recommendations.

sMile

HTML, SCSS, JavaScript, PHP

Web Application

- Winner of ACM (Association for Computing Machinery) WebDex 2024 Hackathon with more than 250+ participants.
- Built a platform addressing 4 Sustainable Development Goals, connecting volunteers with organizations for impactful collaboration.

EXPERIENCE

Stryker AI Research Associate

Jan 2025 – Present

Stryker, Bangalore

- Developed and evaluated computer vision and deep learning pipelines for medical imaging, including image segmentation, image processing, synthetic data generation, and model robustness analysis.
- Worked on 2D/3D medical imaging workflows, leveraging transformer-based architectures, image-to-image translation, 3D synthetic data generation & simulation, and annotation pipelines for AI model development.
- Collaborated with global research and engineering teams on medical imaging datasets, annotation quality assurance, experimental design, and evaluation of AI systems across diverse clinical scenarios.
- Subject to NDA, no further details available.

Machine Learning Researcher

Feb 2024 – Feb 2025

University of Pretoria, South Africa

- Led the Augmentation Team in collaboration with the University of Pretoria's Data Science for Social Good Lab under Prof. Vukosi Marivate.
- Worked on Text augmentation for multilingual NLP models (African languages) with 150,000+ users in the base repository.

MLOps & Data Pipeline Developer

Jun 2024 – Sep 2024

Logeuler Pvt. Ltd.

- Constructed and managed robust data pipelines and workflows to streamline model training, evaluation, and deployment, and monitor model performance within an ERP (Enterprise Resource Planning) environment.

Google Data Science Dept. Lead <i>Google Developer Groups on Campus</i> - Organized a 36-hour national-level hackathon with 5000+ registrations and 250+ participants, driving innovation in real-world tech solutions. - Led a team of 15+ members, organized workshops, and collaborated with an MoU-backed company on real-life crime detection project.	Aug 2024 – Jun 2025 <i>(Promoted)</i>
Office Bearer <i>VITeach (Non-Profit Organization)</i> Effectively managed a diverse team of 250+ volunteers while fostering partnerships with local schools and NGOs to expand outreach and promote education for underprivileged communities.	Sep 2024 – Jul 2025 <i>(Promoted)</i>

PUBLICATIONS

- Joshua Sakthivel Raju, S. Somasundaram, Jaskaran Singh Walia, Srinivas R, and Vukosi Marivate. **"AfroXLMR-Comet: Multilingual Knowledge Distillation with Attention Matching for Low-Resource Languages"**. In *Artificial Intelligence Research. Proceedings of the Southern African Conference for Artificial Intelligence Research (SACAIR 2025), Communications in Computer and Information Science, vol 2784, Springer, Cham. (2026)* [DOI]
- Anbalagan, S., Krishna, H., Joshua S. Raju, et al. **"TamilSTARNet: Segmentation via Tri-phase Architecture and Attention-based Recognition for Handwritten Tamil Characters"**. *npj Heritage Science, vol. 14, 258. (2026)* [DOI]
- Riddhiman Raguraman, Joshua Sakthivel Raju, and Subbulakshimi P. **"Adaptive NPC in Serious Games Using Artificial Intelligence"** (2024) (SSRN Pre-print) (2026) [DOI]
- Joshua S Raju, Krishna, K.V Hitesh, Adityasinh Rana, and Thomas Abraham J.V **"VocalVisageNet: A Spatio-Temporal Transformer approach for Lip-Reading"** (2025) (in press)
- Krishna, K.V Hitesh, Joshua S Raju, Dr. Suganeshwari G, and Dr. Sudha A **"A Shallow ResNet-Based GAN for Super-Resolution Tasks"** (2025) (in press)

ACHIEVEMENTS

Dataset 2024 National-Level Hackathon Winner (1st Place) <i>Nokia, Prodapt, Engagements and Collaborations committee of Data Science</i>	Technical Member 19 Dec 2024 (36hr)
ACM WebDex 2024 Hackathon Winner (1st Place) <i>Association for Computing Machinery</i>	Team Lead 22 Mar 2024 (24hr)
Game-A-Thon Hackathon Winner (1st Place) <i>School of Computer Science and Engineering, VIT, Chennai</i>	Lead Artist 27 Sep 2023

EDUCATION

B.Tech in Computer Science with Specialization in Artificial Intelligence & Machine Learning <i>Vellore Institute of Technology, Chennai, Tamil Nadu</i>	2022 – 2026
Sr. Secondary, CBSE(XII) <i>Delhi Public School, Navi Mumbai, Maharashtra</i>	2020 – 2022
Secondary, CBSE(X) <i>Delhi Public School, Navi Mumbai, Maharashtra</i>	2010 – 2020

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate (1Z0-1122-25)	2025
CISCO: Networking Basics	2024
National Programme on Technology Enhanced Learning: Wild Life Ecology	2024

SPOKEN LANGUAGES

Professional Proficiency :	English, Hindi
Elementary Proficiency :	Telugu, Tamil